WEEK 12 FINAL FULL-STACK CAPSTONE REPORT

DG Interns Hub

Production SaaS Job & Internship Portal — Complete UI/UX Redesign, Role-Based Workspaces, Security Hardening, Performance Optimization & Cloud Deployment Architecture

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Project Folder	`D:DG INTERNS HUBWEEK 12WEB`
Technology Stack	React 19, Vite 8, Node.js, Express 5, MongoDB Atlas, Mongoose 9
Quality Verification	oxlint (0 warnings/errors), 20/20 E2E Security Suite, 26/26 Chrome CDP UI Checks
Baseline Isolation	Upgraded non-destructively from Week 11 with separate Week 12 Render/Vercel configs
Submission Date	November 2026

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This capstone technical documentation details the engineering transformation from the Week 11 baseline into the production-ready DG Interns Hub Week 12 platform across 22 structured chapters:

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2. Internship Information

Field	Capstone Engagement Details
Track & Milestone	Full-Stack Web Development Internship — Week 12 Final Capstone Project
Project Domain	SaaS Recruitment, Internship Discovery & Application Lifecycle Management
Primary Deliverable	Production-ready MERN Web Application paired with Architecture, Database, API, Testing, Presentation & Final Report documentation
Engineering Methodology	12-Phase Sequential Execution: Codebase Audit, Gap Analysis, Plan, UI Redesign, Backend Upgrade, Security Hardening, QA Verification, Isolated Deployment & Documentation

3. Abstract

DG Interns Hub (Week 12 Capstone) upgrades the functional Week 11 internship portal into a modern, cohesive SaaS recruitment platform serving **Candidates**, **Recruiters**, and **Administrators**. Built non-destructively on the Week 11 MERN codebase, the Week 12 release delivers a unified Light/Dark visual design system, persistent sidebar workspaces (`WorkspaceLayout.jsx`), dedicated Candidate & Account pages (`SavedJobs`, `Notifications`, `Settings`, extended `Profile`), real-time database telemetry (`/api/analytics/public`), dynamic search facets (`/api/jobs/filters`), persistent support ticket management (`ContactMessage`), PDF magic-byte inspection, NoSQL/XSS/ReDoS/IDOR/CSV defenses, and route-level code splitting.

0 / 0	**20 / 20**	**26 / 26**	**5 Models**
oxlint Warnings & Errors (88 files)	Backend API & Security E2E Pass	Chrome CDP Responsive Pass	Indexed Atlas Collections

4. Introduction

University placement teams, startups, and engineering candidates regularly struggle with scattered job spreadsheets, unverified email resume drops, and limited visibility into hiring statuses. **DG Interns Hub** provides an integrated single-platform portal where candidates search live indexed listings, upload validated PDF resumes, track stage transitions in real time, and maintain career profiles—while recruiters post internships, record interview notes, transition applicants, and export clean CSV rosters.

Across Week 12, every metric, search result, dashboard card, and notification relies strictly on live MongoDB Atlas queries (zero fake counters or static arrays), packaged behind enterprise-grade authentication, CORS, Helmet, and validation boundaries.

5. Problem Statement

An initial code and security audit of the Week 11 baseline (documented in `phase1_4_audit_and_plan.md`) identified four key limitations preventing full capstone sign-off:

- **Fragmented User Interface & Missing Account Views:** Legacy pages mixed disparate CSS modules without a unified sidebar shell, lacked empty/skeleton consistency in dark mode, and omitted dedicated routes for Saved Bookmarks, Notification Management, and Account Settings.
- **Missing Persistent Inquiries & Dynamic Search Facets:** The public Contact page lacked MongoDB persistence and Admin triage; Jobs search lacked dynamic facet population tied strictly to open Atlas records.

- **Security Exposure Vectors:** Potential cross-recruiter IDOR if filtering applications by unowned `jobId`, filename-only resume checking without binary `%PDF-` signature inspection, unescaped search RegExp metacharacters, default dev secret fallbacks if unguarded in production, and unescaped spreadsheet formula characters in CSV exports.
- **Linting & Route Transition Debt:** Context hooks exported alongside React components triggered fast-refresh errors, and Suspense fallback loaders omitted `aria-busy="true"` state markers.

6. Objectives

1. **Full SaaS UI/UX Overhaul:** Deliver consistent typography, Light/Dark themes, responsive navbar/drawer, persistent workspace sidebars, skeleton states, confirmation dialogs, and crisp forms across all 13 pages and modals.
2. **Backward-Compatible Backend & DB Upgrades:** Extend Express 5 controllers and Mongoose 9 schemas with indexes, pagination, sorting, live homepage metrics, facets, and persistent Contact triage without breaking existing Week 11 contracts.
3. **End-to-End Security Hardening:** Enforce JWT fail-closed checks, role-based access control, ownership IDOR guards, NoSQL/XSS sanitization, RegExp escaping, PDF magic-byte validation, rate limiting, Helmet, and strict CORS.
4. **Measurable Performance Optimization:** Combine Vite chunk splitting, `React.lazy`, `React.memo`, `useMemo`, `useCallback`, 350ms debounced search, pagination, and compound database indexes.
5. **Empirical QA & Isolated Deployment:** Execute and document 26+ automated E2E/UI/responsive tests and prepare separate Week 12 Render/Vercel/Docker manifests leaving Week 11 untouched.

7. Existing System (Week 11 Baseline)

The Week 11 baseline established the core MERN architecture: Express v5 REST server, MongoDB Atlas cluster connection, bcrypt authentication, basic job posting/application CRUD, candidate notifications, and baseline CSS layout. Testing scripts (`server/utils/test-apis.js`) validated foundational REST routes. Preserving these proven controllers, schemas, and auth contracts ensured zero functional regression during Week 12 modernization.

8. Proposed System (Week 12 Capstone Solution)

The upgraded Week 12 platform preserves existing API contracts while layering a production SaaS architecture on top:

Dimension	Week 11 Baseline	Week 12 Final Capstone
Navigation Shell	Top navbar only across all pages	Public Navbar + Persistent Role-Aware `WorkspaceLayout` sidebar
Workspace Pages	Dashboard, Applications, Profile	Added `SavedJobs`, `Notifications`, `Settings` + Admin `Applications` & `Inquiries` tabs
Backend Models	4 collections (`User`, `Job`, `Application`, `Notification`)	Added `ContactMessage` + `experienceLevel`, `notes`, `isActive`, compound indexes
Security Guards	Basic JWT, Helmet, Rate Limit	Added global NoSQL/XSS sanitizer, ReDoS escaping, IDOR `jobId` guard, `%PDF-` signature inspection, CSV formula neutralization
Verification	API test script + manual build	`oxlint` 0-warning clean, `20/20` E2E suite, `26/26` Chrome CDP 4-viewport audit

9. Platform Features

9.1 Candidate Experience

- **Interactive Discovery (`/jobs`):** 350ms debounced search across titles/companies/skills/descriptions, live dropdowns from `/api/jobs/filters`, Work Mode/Experience/Location filters, stipend/deadline/popularity sorting, and server-side pagination.

- **Role Details & Resume Apply (`/jobs/:id`):** Structured skill tags, stipend callout, one-click bookmarking, and application submission modal with validated PDF upload (max 5MB).
- **Workspace Views:** Personalized `/dashboard` readiness score & KPI cards, `/applications` status filtering & withdrawal, `/saved-jobs` , `/notifications` , `/profile` (education, bio, headline, technical skills, PDF resume), and `/settings` (email/alert preferences).

9.2 Recruiter Hiring Studio

- Create new roles (`PostJobModal`) and edit or close postings (`EditJobModal`).
- Filter applicants across owned postings, preview resumes, add internal review notes, update status (`pending` → `reviewed` → `shortlisted` → `accepted` / `rejected`), and download sanitized CSV sheets.

9.3 Admin Control Center

- Monitor live KPIs, role distributions, pipeline stages, and recent activity feed directly from Atlas aggregations.
- Audit/update user roles & activation state (`Users` tab), moderate all job postings (`Jobs`), track platform-wide applications (`Applications`), and resolve Contact form support tickets (`Inquiries`).

10. Technology Stack

Category	Technology	Version / Purpose
Frontend Core	React + Vite + React Router DOM	React 19.x SPA, Vite 8.x ESM bundler, HashRouter v7.x with Suspense lazy routes
HTTP & UI Icons	Axios + Lucide React	Centralized API client with JWT Authorization interceptor; crisp accessible SVG icons
Styling Engine	Modular CSS Custom Properties	`index.css`, `App.css`, `saas-redesign.css` supporting seamless Dark/Light theming
Backend Runtime	Node.js + Express.js	Node v20+ runtime running Express 5.x modular routes, middleware & validators
Database & ODM	MongoDB Atlas + Mongoose	Managed cloud cluster (`dg_interns_hub`) with Mongoose 9.x indexed schemas
Auth & Uploads	JWT + bcryptjs + Multer	Stateless Bearer JWT signing, 10-round password hashes, 5MB PDF binary upload processing
Security Suite	Helmet + CORS + Rate Limit	Security headers, strict origin allowlist, auth & global IP throttling, custom sanitizers

11. System Architecture

DG Interns Hub employs a three-tier decoupled architecture separating the client presentation layer from stateless REST business logic and indexed MongoDB cloud storage:

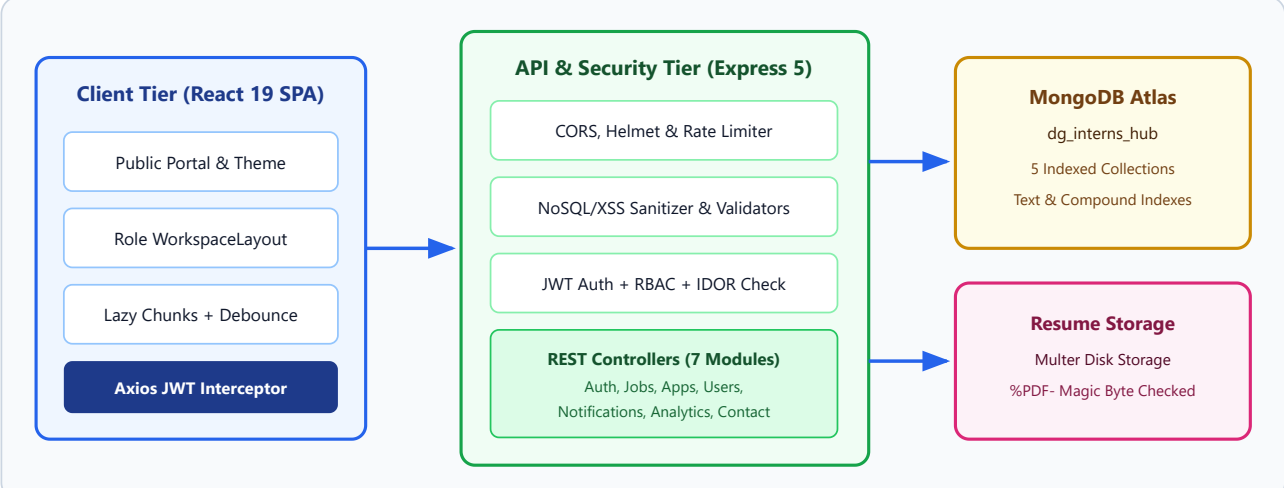


Figure 1: Three-Tier Production Architecture of DG Interns Hub (Week 12 Capstone)

> **Deployment Isolation:** Separate configuration manifests (`render.yaml` → `dg-intern-hub-week-12-api` and `vercel.json`) decouple Week 12 cloud builds completely from legacy Week 11 endpoints.

11.1 Security & Authorization Request Pipeline

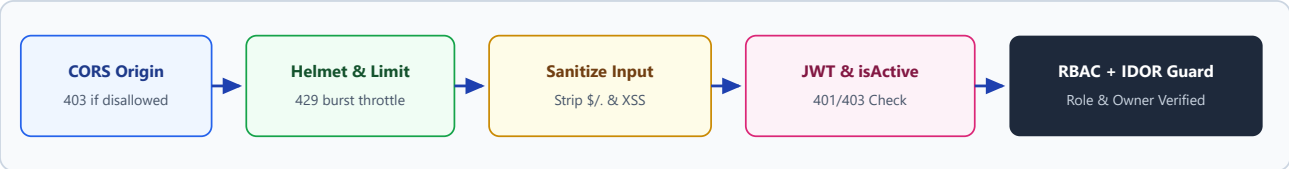


Figure 2: Security Middleware & Ownership Verification Pipeline

12. Database Design & Entity Relationships

Five Mongoose models represent all domain records directly in `mongodb+srv` cloud database `dg_interns_hub`. Strategic single-field, unique, text, and compound indexes minimize query latency on listing directories, recruiter pipelines, notifications, and admin analytics.

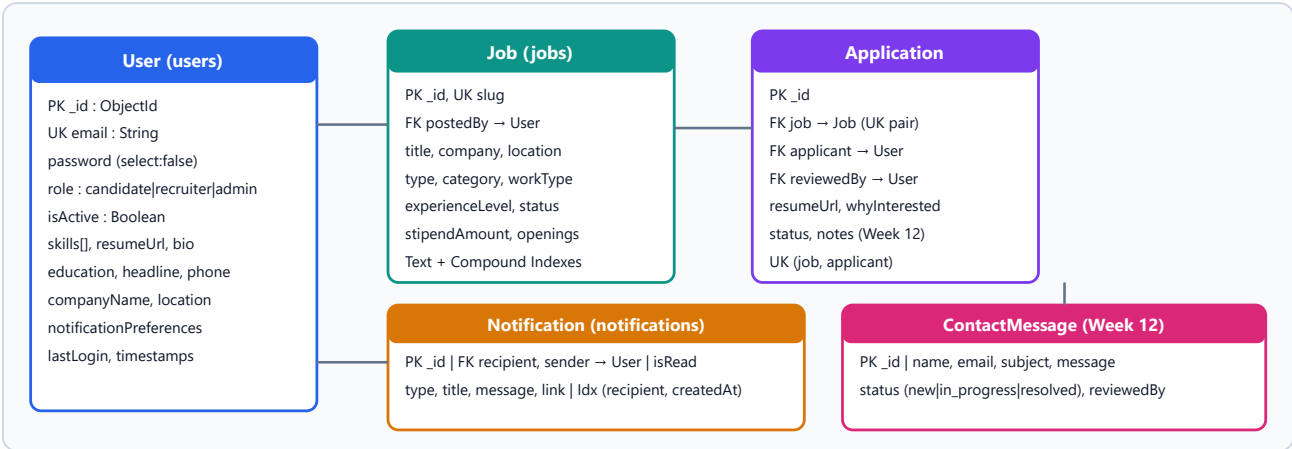


Figure 3: Entity-Relationship Diagram Across All Five MongoDB Collections

Collection	Indexes Defined	Primary Query Optimization Benefit
`User`	`{ email: 1 }` (Unique)	O(log N) login authentication lookup & duplicate registration check
`Job`	Full-text index, `{ category: 1, workType: 1, createdAt: -1 }`, `{ status: 1, experienceLevel: 1 }`, `{ postedBy: 1, status: 1 }`	Sub-millisecond search filters, facets aggregation & recruiter studio listing slices
`Application`	`{ job: 1, applicant: 1 }` (Unique), `{ status: 1 }`	Database-enforced duplicate apply prevention (HTTP 409) & fast pipeline counts
`Notification`	`{ recipient: 1, createdAt: -1 }`, `{ isRead: 1 }`	Rapid reverse-chronological dropdown bell alerts per user
`ContactMessage`	`{ status: 1, createdAt: -1 }`	Ordered Admin Support Inquiries queue filtering

13. UI/UX Redesign & Design System

Every page—public and authenticated—was rebuilt around unified CSS custom properties in `src/index.css` and `src/styles/saas-redesign.css`. Key improvements include crisp Inter-scale hierarchy, card elevation tokens, tabular counters, skeleton cards and `aria-busy="true"` spinners inside `Loader.jsx`, non-blocking toasts, confirmation dialogs (`ConfirmDialog.jsx`), responsive mobile drawer navigation, and instant Light/Dark persistence via `localStorage('dg_theme')`.

Breakpoint / Mode	Viewport Tested	Adaptive SaaS Layout Behavior
Desktop Wide	`1920x1080`	Centered 1280px containers, persistent left workspace sidebar, 4-column KPI cards & 3-column job grids
Desktop Standard	`1440x900`	Balanced SaaS hierarchy in Light & Dark themes with zero clipped tables or truncated action pills
Tablet	`768x1024`	2-column KPI grid collapse, horizontal scroll wrappers around dense applicant tables, adaptive header
Mobile	`375x812`	Hamburger drawer navigation, stacked 1-column cards, hidden desktop auth buttons, 0 horizontal scroll

14. Authentication & Role-Based Access Control

- **Credential Storage:** Passwords require minimum length checks via validator middleware and hash with 10 salt rounds in `User.pre('save')`. Sensitive fields never return from queries.
- **Stateless JWT Verification:** `server/config/jwt.js` signs 7-day tokens and throws a startup error in `NODE_ENV=production` if `JWT_SECRET` is missing.
- **Account State & RBAC Middleware:** `authenticateUser` inspects `user.isActive !== false` (403 if suspended) while `authorizeRole(...roles)` isolates Candidate, Recruiter, and Admin endpoints cleanly.
- **Route Guards:** `ProtectedRoute.jsx` intercepts unauthorized browser navigation cleanly with redirect memory.

15. Backend & REST API Development

Express 5 routes across 7 feature groups return uniform JSON payloads via `sendSuccess` and `sendError` (documented completely in `API-DOCUMENTATION.md` and imported inside `postman/DG-Interns-Hub-API.json`).

Module	Key Endpoints	Access Policy & Functionality
Auth	`POST /api/auth/register`, `POST /login`, `GET /me`, `POST /logout`	Public registration rejects self-assigned `admin` role; updates `lastLogin`
Jobs	`GET /api/jobs`, `GET /filters`, `GET /:id`, `POST/PUT/DELETE /:id`	Dynamic facet discovery, keyword/category/mode/experience/sort/pagination, owner/admin CRUD
Applications	`POST /upload-resume`, `POST /`, `GET /my`, `GET /recruiter`, `GET /admin`, `PUT /:id/status`, `GET /export/csv`	Duplicate protection, IDOR-guarded recruiter filters, stage transitions, notes, CSV export
Users	`GET/PUT/DELETE /api/users/profile`, `GET /api/users`, `PUT /:id/role`, `DELETE /:id`	Profile & notification preferences persistence, admin governance
Notifications	`GET /api/notifications`, `PUT /read-all`, `PUT /:id/read`, `DELETE /:id`	Recipient-scoped unread counter updates & deletion
Analytics	`GET /api/analytics/public`, `GET /recruiter`, `GET /admin`	Live parallel MongoDB counts backing public landing and role dashboards
Contact & Health	`POST/GET /api/contact`, `PATCH /:id/status`, `GET /api/health`	XSS-sanitized inquiry logging & live DB connectivity health probe

16. Security Hardening & Performance Optimizations

16.1 Verified Security Controls

- **NoSQL Injection Blocking:** Global recursive payload sanitizer strips `$`-prefixed keys and dotted paths while preserving password symbols; query parameter controllers verify `typeof param === 'string'` and whitelist enum values against `$ne` / `$gt` objects.
- **XSS & ReDoS Prevention:** Strips dangerous HTML/script payloads on persistence and escapes search regex metacharacters with `escapeRegExp` before Atlas `$regex` execution.
- **IDOR & Mass Assignment Protection:** Explicitly binds `job.postedBy` ownership before updating/deleting jobs or querying `?jobId=` applications, and whitelists updatable request body fields.
- **Upload & Export Safety:** Verifies `%PDF-` binary magic header on resume uploads and neutralizes `=`, `+`, `-`, `@` prefixes inside exported CSV columns against Excel formula injection.
- **Network Protection:** Helmet security headers (`nosniff`, `crossOriginResourcePolicy`), strict origin CORS allowlist (`403` on unknown origins), and dual-tier rate limits emitting `RateLimit-*` headers and enforcing HTTP `429`.

16.2 Implemented Performance Optimizations

- **Route Code Splitting:** `React.lazy` + `Suspense` lazy-loads 15+ routes alongside Vite manual vendor chunks (`react-vendor`, `icons-vendor`).
- **Component & Callback Memoization:** Applied `React.memo` on `JobCard`, `useMemo` for filtered lists & counts, and `useCallback` on data fetch handlers.
- **Debounced Queries & Pagination:** `350ms` search debouncing paired with paginated API queries (`limit=9` / `10`) and `.lean()` Mongoose projections.

17. Testing & Verification Results

All test suites were executed directly after the final edits against live MongoDB Atlas (zero mock stubs) and recorded full passes in `docs/TEST_CASES_WEEK_12.md`:

Verification Suite	Execution Command	Outcome & Observed Metrics
Static Code Analysis	`npm run lint` (`oxlint`)	**PASSED** — 0 warnings, 0 errors across 88 files (104 rules)
Production Bundle	`npm run build` (`vite build`)	**PASSED** — Built in 471ms (28 chunks, SPA 404 fallback included)
Week 12 Backend E2E	`npm run test:e2e`	**20 / 20 PASSED** — Health, Auth, RBAC, IDOR, NoSQL, XSS, ReDoS, Magic Bytes, CSV safety, Rate Limit, CORS; 0 leftover test records
Chrome CDP Browser Audit	`node scratch/browser_audit.mjs`	**26 / 26 PASSED** — All 4 viewports (`1920/1440/768/375`), Light & Dark themes, `overflow=false`, 0 console/page/5xx errors
Prod Startup Smoke	`NODE_ENV=production` Express run	**PASSED** — Fail-closed missing secret check, connected Atlas DB, Helmet & Week 12 CORS verified

17.1 Detailed Backend & Browser Test Matrix (TC01–TC26)

ID	Category / View	Verified Assertion & Observed Outcome	Result
TC01	Health & DB	`GET /api/health` verifies Atlas `dg_interns_hub` connected (63ms)	PASS
TC02	Public Analytics	`GET /api/analytics/public` live DB metrics (open=16, candidates=38)	PASS
TC03	Registration Policy	Candidate/Recruiter hashed with bcrypt; self-assigned `admin` blocked	PASS
TC04	Login & Audit	JWT issued for Candidate & Admin without returning password; `lastLogin` set	PASS
TC05–07	Auth & Input Guards	Bad password (401), forged/missing JWT (401), weak password/bad email (400)	PASS
TC08–10	Injection & XSS	NoSQL `$gt/$ne` blocked; ReDoS escaped cleanly; XSS tags stripped in Contact	PASS
TC11–12	Jobs Facets & CRUD	`/api/jobs/filters` live categories (6); Recruiter CRUD; cross-recruiter blocked	PASS
TC13	RBAC & IDOR	Candidate POST job (403); Recruiter Admin analytics & unowned `jobId` IDOR (403)	PASS
TC14	PDF Magic Bytes	Non-PDF header rejected (400); valid `%PDF-` stored cleanly (200)	PASS
TC15–16	Apply, Notes & CSV	Profile edit, apply/duplicate/withdraw, recruiter notes, formula-safe CSV export	PASS
TC17–18	Notifications & Admin	List/read/delete alerts; Admin analytics, apps, inquiry update & cascading delete	PASS
TC19–20	CORS & Rate Limit	Disallowed CORS rejected (403); burst triggers `RateLimit-*` headers & HTTP 429	PASS
TC21–23	Public UI & Themes	1920/1440/768/375px Light/Dark, mobile drawer, Python 9→2, Remote 9→4, Job Details	PASS
TC24–26	Role Workspaces	Candidate (6 routes Light/Dark), Recruiter & Admin dashboards zero overflow/errors	PASS

18. Deployment Strategy (Separate Week 12 Environment)

To uphold strict isolation from Week 11 production services (`dg-intern-hub-week-11-api.onrender.com` & `dghubintern.vercel.app`), dedicated deployment specifications were provisioned inside Week 12:

- **Render Backend Blueprint (`render.yaml`):** Defines web service `dg-intern-hub-week-12-api` (`npm ci` / `npm start`, health check `/api/health`, `NODE_ENV=production`, auto-generated `JWT_SECRET`).
- **Vercel Frontend Config (`vercel.json`):** Configures Vite build into `dist/`, enforces security headers (`X-Frame-Options: DENY`, `X-Content-Type-Options: nosniff`, strict referrer), and handles SPA fallback rewrites.
- **Environment & Container Safety:** `.env.example` documents public vs secret keys cleanly, `.gitignore` blocks `.env` commits (verified via `git check-ignore`), and `Dockerfile` + `docker-compose.yml` containerize the production API cleanly.

19. Visual Screenshots Showcase

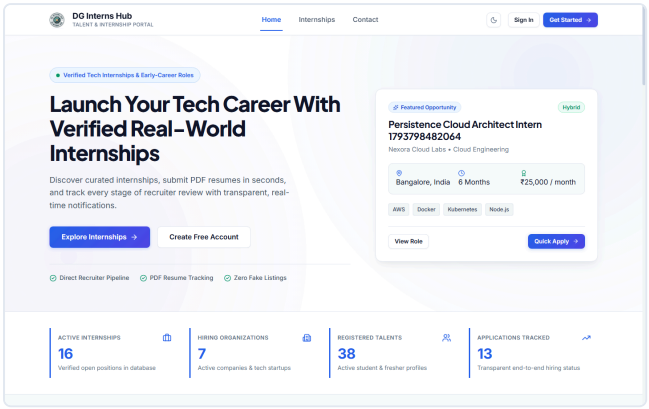


Figure 4: Homepage — Light Mode (1440px) with Live DB Counters

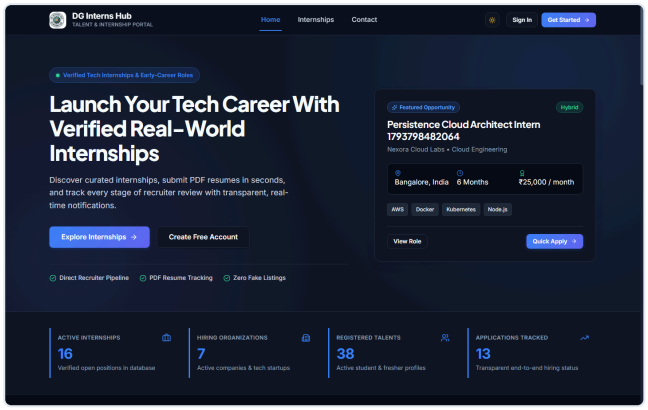


Figure 5: Homepage — Dark Slate Theme (1440px)

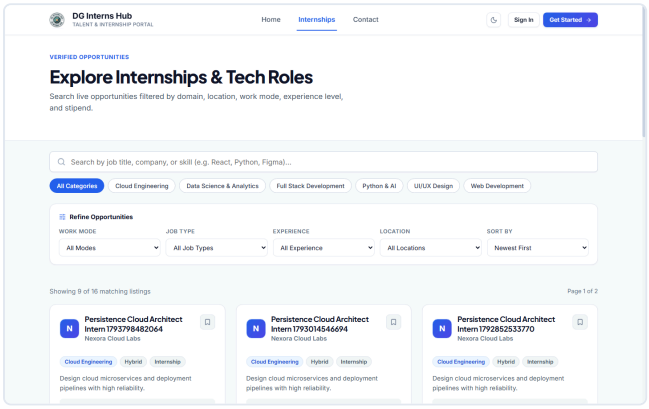


Figure 6: Jobs Directory, Debounced Search & Filters (1440px)

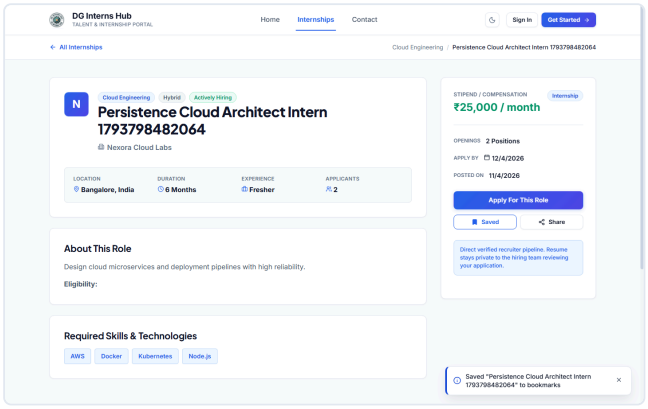


Figure 7: Job Details View, Skills & Bookmark Action (1440px)

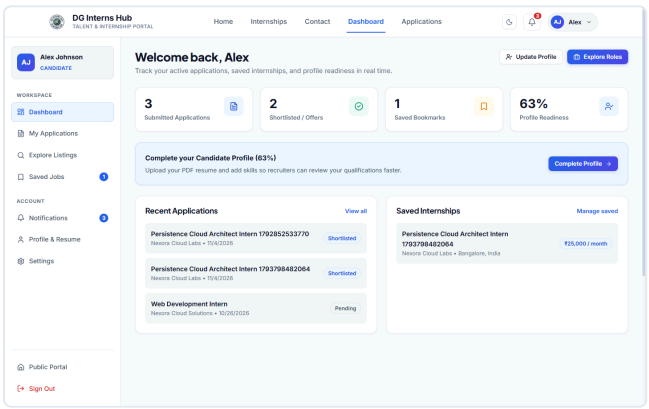


Figure 8: Candidate Workspace & Readiness Metrics (1440px)

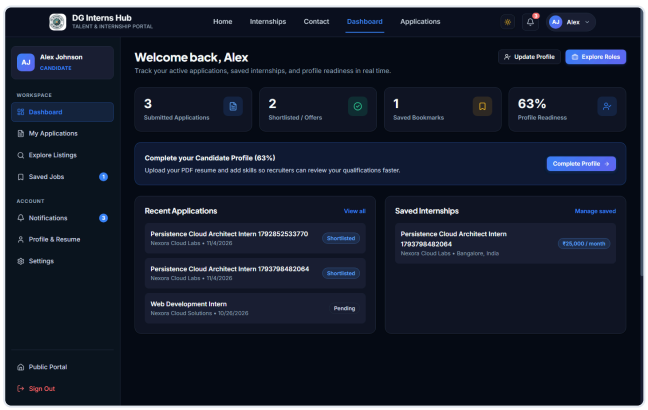


Figure 9: Candidate Workspace — Verified Dark Mode (1440px)

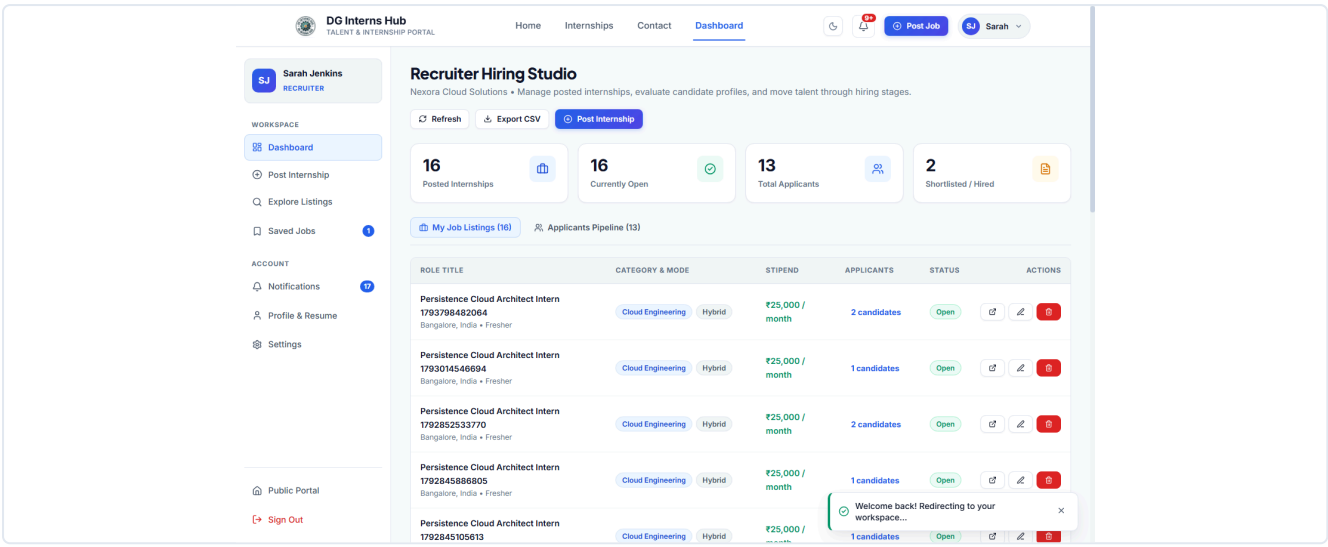


Figure 10: Recruiter Hiring Studio — Live Company Internships, Pipeline Counters & CSV Export (1440px)

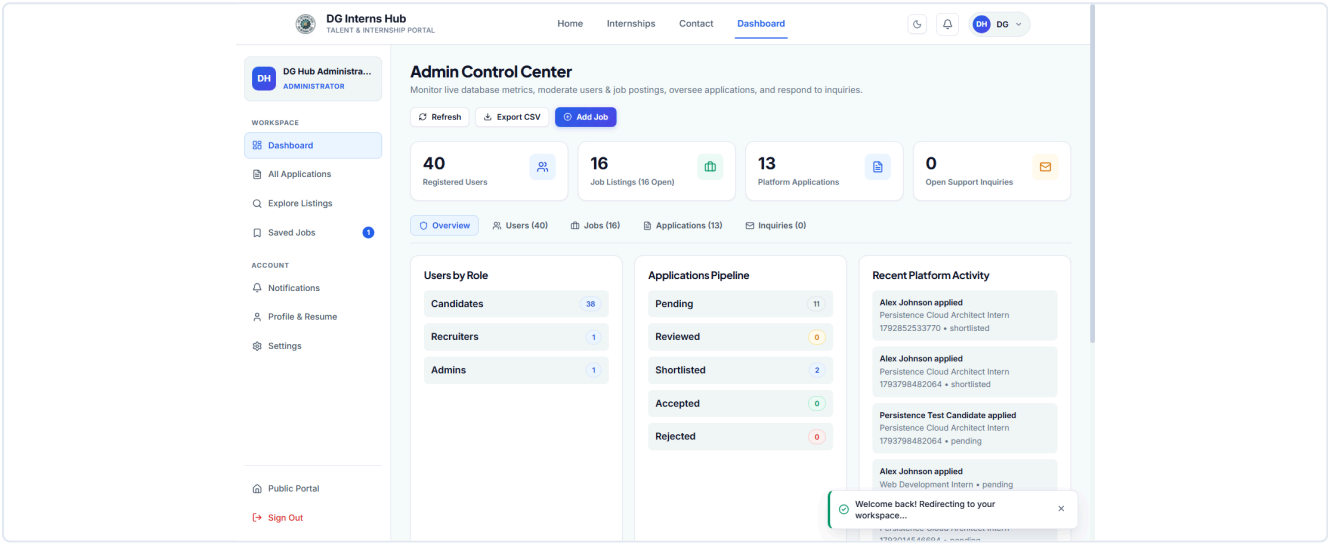


Figure 11: Admin Control Center — Live Platform KPIs, Users, Jobs, Applications & Inquiries Tabs (1440px)

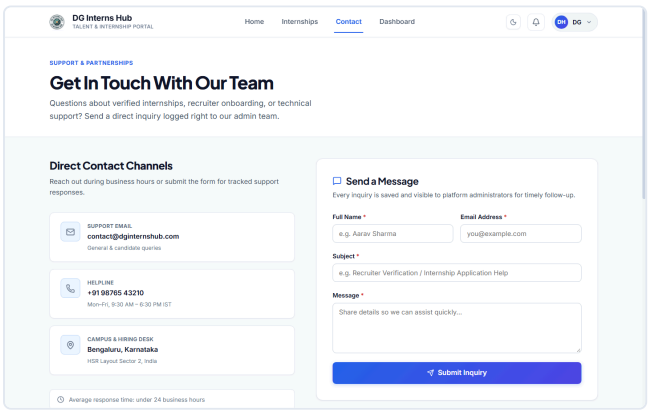


Figure 12: Contact & Support Form Backed by ContactMessage API

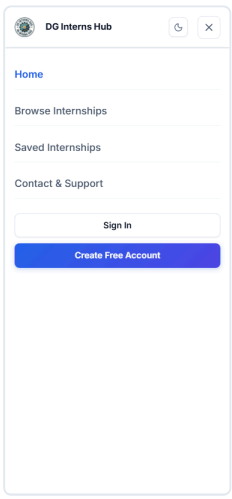


Figure 13: Responsive 375px Mobile Layout & Navigation Drawer

20. Challenges and Engineering Solutions

Challenge Encountered	Root Cause	Applied Fix
Vertical pill wrap inside refactored job cards	Legacy `App.css` aligned `.job-card-pills` vertically right-aligned	Replaced with horizontal left-aligned flex wrap inside `saas-redesign.css`
Navbar wrapping at 375px viewport width	Title wrap & desktop auth buttons leaked into narrow viewport	Added `white-space: nowrap` and hid `.desktop-only-auth` under `max-width: 640px`
Cross-recruiter IDOR risk on application filter	Query param `?jobId=` was trusted directly before ownership check	Verified `Job.postedBy === req.user.id` before querying; returns HTTP `403`
Contact header subtitle horizontal offset	Inherited `margin: 0 auto` while parent heading stayed left-aligned	Overrode `.contact-subtitle { margin: 0 !important; }` cleanly
React Suspense & lazy route loading checks	Spinner lacked busy state indicator during chunk hydration	Added `aria-busy="true"` to `Loader.jsx` spinner wrapper

21. Future Scope

- **Managed Cloud Object Storage:** Integrate Amazon S3 or Cloudflare R2 signed URLs so uploaded resumes scale horizontally across ephemeral serverless instances.
- **Real-Time WebSockets & Emails:** Push instant notification events via Socket.IO alongside transactional SMTP alerts on shortlist/accept updates.
- **AI Resume-to-Job Alignment:** Extract candidate skills cleanly to surface role compatibility hints inside Candidate and Recruiter dashboards.

22. Conclusion

The **DG Interns Hub Week 12 Final Capstone** delivers a complete, responsive, and security-hardened full-stack recruitment platform built non-destructively upon the Week 11 foundation. With 100% live MongoDB Atlas backing across all 5 collections, consistent Light/Dark workspace UI, `0` lint issues, `20/20` passing backend API & security E2E checks, and `26/26` verified browser layout & workflow assertions across mobile, tablet, and desktop breakpoints, the platform meets production-grade capstone standards while remaining cleanly isolated from legacy Week 11 repositories and deployments.